

Renan Souza

✉ contact@renansouza.org • 🌐 RenanSouza.org • in renansouza1
🔑 ID: x9t36ewAAAAJ • 🌐 renan-souza • 🆔 0000-0002-1794-808X

Updated on October 16, 2025.

Bio

Renan Souza earned his Ph.D., M.Sc., and B.Sc. in Computer Science (2009–2019) from the Federal University of Rio de Janeiro (UFRJ). Since 2022, he has been a researcher and software engineer at Oak Ridge National Laboratory, after spending seven years at IBM. He was a visiting scientist at INRIA, France, during his Ph.D. and, during his B.Sc., studied abroad at Missouri State University and interned at SLAC National Laboratory (Stanford University). Active in engineering, research, and technical leadership since 2010, he has authored 50+ peer-reviewed papers in leading venues and holds 10+ USPTO patents. His work has influenced researchers, industry, and practitioners in fields of AI, distributed systems, and data management across diverse application domains. His current focus is on designing and building scalable systems to support responsible and trustworthy AI workflows.

Research Interests

Large-scale Data Science and Data Engineering • Edge–Cloud–HPC Workflows • AI Workflows • Provenance Data • Machine Learning Systems • Agentic AI

Education

- **Ph.D. in Computer Science**, Federal Univ. of Rio de Janeiro, Brazil Sep 2015 – Dec 2019
Supervised by Marta Mattoso (COPPE/UFRJ) and Patrick Valduriez (Inria).
Title: Supporting User Steering in Large-scale Workflows with Provenance Data
- Visiting Ph.D. Student, Inria/Univ. Montpellier, France Jan 2019 – Mar 2019
Supervised by Patrick Valduriez (Inria).
- **M.Sc. in Computer Science**, Federal Univ. of Rio de Janeiro, Brazil Jan 2013 – Jul 2015
Supervised by Marta Mattoso (COPPE/UFRJ).
Title: Controlling the Parallel Execution of Workflows Relying on a Distributed Database
- Computer Science exchange student, Missouri State University, U.S. Jun 2011 – Jun 2012
- **B.Sc. in Computer Science**, Federal Univ. of Rio de Janeiro, Brazil Jan 2009 – Dec 2012
Supervised by Maria Luiza Machado Campos (DCC/UFRJ).
Title: Linked Open Data Publication Strategies: An Application in Network Performance Data
- **Technical Degree in Information Systems**, Lemos de Castro Jan 2005 – Dec 2007

Experience

- **Oak Ridge National Laboratory** Oct 2022 – Present
Research Scientist, HPC Workflows, Data & AI Knoxville, United States
As Principal Investigator and lead software developer, he leads the design and development of large-scale data and AI systems within the Workflows and Ecosystem Services group at ORNL, focusing on AI-driven observability, data integration, workflow provenance, and LLM-based agentic workflows to accelerate scientific discovery in Edge–Cloud–HPC environments.

- **IBM Research**
Research Scientist and Software Eng., Cloud, Data & AI

As a Staff Research Scientist (2021–2022), he was a lead researcher and developer on projects in large-scale data science and engineering to support AI systems in hybrid cloud and cluster environments with highly distributed and heterogeneous workloads for clients across energy, finance, physics, and cheminformatics domains. Although primarily focused on backend development, data engineering, cloud, and DevOps, he collaborated closely with front-end developers and HCI researchers to design domain-specific applications. As a Research Software Engineer (2015–2021), he worked with and led R&D projects on large-scale data integration for AI systems in the energy sector, targeting cluster and cloud platforms. He also led the Cloud DevOps team responsible for developing and deploying conversational AI systems.

As a Software Engineering Intern (2015), he designed and implemented big data and machine learning solutions for real-time analysis of streaming social data.

Apr 2015 – Oct 2022
Rio de Janeiro, Brazil
- **SLAC National Accelerator Laboratory, Stanford Univ.**
Research Software Engineering intern

Led the development of a cloud platform utilizing semantic web, big data, and data warehousing techniques. This platform is designed to store, retrieve, visualize, and publish structured data about internet performance worldwide, enabling understanding of global Internet quality.

May 2013 – Dec 2014
Menlo Park, United States
- **COPPE-UFRJ**
Software Engineer

As a Lead Software Engineer (2013–2014), he led the development of a system that facilitated access of information about public services offered by the Brazilian Federal Government. Also lead the development of a platform to publish linked open data from the Brazilian Federal Register on the semantic web, applying agile practices, ontology-based data modeling, and natural language processing.

As a Full-Stack Software Engineering Intern (2011–2013), he worked on the R&D of various web systems.

Dec 2011 – Sep 2014
Rio de Janeiro, Brazil
- **Federal Univ. of Rio de Janeiro**
Software Engineering intern

Developed a system that integrated data warehouse environments with both structured and unstructured data, enabling the generation of more intelligent and flexible information reports.

Jan 2010 – Jul 2011
Rio de Janeiro, Brazil
- **Petrobras**
IT Intern

Implemented features and provided ongoing maintenance for web systems supporting Petrobras employees.

May 2007 – May 2008
Rio de Janeiro, Brazil

Technical Knowledge

- **Languages:** Python, Java, Shell scripting, C, C++, C#, NodeJS, Scala, Lua
- **Data Science/ML Technologies:** Pandas, Polars, Jupyter Notebooks, Numpy, Matplotlib, Seaborn, Plotly, Tensorflow, ScikitLearn, Keras, PyTorch, MLFlow, Airflow, Grafana
- **Agentic AI and LLMs:** MCP Agents, Crew AI (Multi-agent Framework), LangChain, Streamlit for AI Agents, RAG, Prompt Engineering Techniques
- **Big Data & Parallel Processing Frameworks:** Dask; Apache Spark: RDD, DataFrames, Streaming, MLib, GraphX, GraphFrames; Hadoop Ecosystem
- **Cloud and Cluster computing:** VMs, Dockers, Kubernetes, OpenShift, HPC (Slurm, LSF, PBS)
- **DevOps:** Containers, Kubernetes, OpenShift, CI/CD Pipelines, GitHub, GitHub Actions, Travis, Jenkins
- **Message Queueing Systems:** Kafka, RabbitMQ, Redis
- **GPU Programming and Profiling:** NVIDIA and AMD Python APIs for GPU performance analysis
- **Relational DBMS:** PostgreSQL/PostGIS, DB2, SQLite, MySQL, MySQL Cluster, MS SQL Server
- **NoSQL DBMS:** MongoDB, AllegroGraph, Jena, Blazegraph, Virtuoso, Sesame, Cloudant, CouchBase, Redis, Impala, Elasticsearch, HBase, Hive, Apache Ignite, LMDB
- **Heterogeneous Data Management:** Data Integration, Multi-database Queries, Polystores
- **Cluster Deployment:** YARN, Mesos, Standalone deployment
- **Business Intelligence:** MS SQL Server BI developer studio, Pentaho Solutions, Talend;
- **Semantic Web Tools/Languages:** OWL, RDF, SPARQL, Protege

- **Distributed and Concurrent Programming:** PubSub, MPI, OpenMP, CUDA, Data-centric distributed and parallel programming
- **Web Development:** Python Flask/UWSGI, Java EE, Tomcat/JBoss, Spring Boot

Selected Publications

For complete list, visit: RenanSouza.org/publications

- [1] **R. Souza**, T. J. Skluzacek, S. R. Wilkinson, M. Ziatdinov, R. F. Silva, "Towards lightweight data integration using multi-workflow provenance and data observability," in *IEEE International Conference on e-Science*, 2023. DOI: 10.1109/e-Science58273.2023.10254822. [Online]. Available: <https://doi.org/10.1109/e-Science58273.2023.10254822>.
- [2] **R. Souza**, L. G. Azevedo, V. Lourenço, E. Soares, R. Thiago, R. Brandão, D. Civitarese, E. Vital Brazil, M. Moreno, P. Valduriez, M. Mattoso, R. Cerqueira, M. A. S. Netto, "Workflow provenance in the lifecycle of scientific machine learning," *Concurrency and Computation: Practice and Experience*, vol. e6544, pp. 1–21, 2021. DOI: 10.1002/cpe.6544. [Online]. Available: <https://doi.org/10.1002/cpe.6544>.
- [3] **R. Souza**, A. Gueroudji, S. DeWitt, D. Rosendo, T. Ghosal, R. Ross, P. Balaprakash, R. F. Silva, "Prov-agent: Unified provenance for tracking AI agent interactions in agentic workflows," in *IEEE International Conference on e-Science*, Chicago, U.S.A.: IEEE, 2025.
- [4] **R. Souza**, S. Caino-Lores, M. Coletti, T. J. Skluzacek, A. Costan, F. Suter, M. Mattoso, R. F. Silva, "Workflow provenance in the computing continuum for responsible, trustworthy, and energy-efficient AI," in *IEEE International Conference on e-Science*, Osaka, Japan: IEEE, 2024. DOI: <https://doi.org/10.1109/e-Science62913.2024.10678731>.

Grants and Awards

- 2nd IBM Patent Plateau (8+ patents submitted to USPTO) 2021
- SBBD Honored Mention for the Best Ph.D. Thesis Award 2021
- 1st IBM Patent Plateau (4+ patents submitted to USPTO) 2020
- SBBD Best M.Sc. Thesis Award 2017
- SBBD Honored Mention on the paper
Spark Scalability Analysis in a Scientific Workflow 2017
- CAPES M.Sc. Grant 2013 – 2014
- Brazil Science Mobility Grant - Missouri State University 2012 – 2013
- Scientific Initiation Grant - Federal Univ. of Rio de Janeiro 2010

Teaching and Supervisions

Teaching:

- Databases Laboratory, graduate, UFRJ 2017
Teacher assistant to Prof. Marta Mattoso
- Logics for Computer Science, undergraduate, UFRJ 2012–2013
Teacher assistant to Prof. Mario Benevides

Supervisions of final dissertations

- Pedro Paiva Miranda, undergraduate, UFRJ, Co-supervision with Prof. Marta Mattoso 2015
Thesis title: *A Mechanism for Fault Tolerance in Parallel Executions of Workflows supported by a Database*
- Rachel Gonçalves de Castro, undergraduate, UFRJ, Co-supervision with Prof. Marta Mattoso 2015
Thesis title: *Publication of Workflow Provenance Data in the Semantic Web*

Talks and Participation in Events

- **IEEE/ACM Supercomputing (SC)** in St. Louis, U.S.A. 2025

- LLM Agents for Interactive Workflow Provenance: Reference Architecture and Evaluation Methodology, Oral presentation , link
- **IEEE International Conference on e-Science** in Chicago, U.S.A 2025
 - Large-scale Workflow Provenance Data Management in the AI Lifecycle using Flowcept, Tutorial , link
 - PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows, Oral presentation , link
- **IEEE International Conference on e-Science** in Osaka, Japan (Virtual) 2024
 - Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI, Oral presentation , link
- **IEEE/ACM Supercomputing (SC)** in Atlanta, GA 2024
 - Integrating Evolutionary Algorithms with Distributed Deep Learning for Optimizing Hyperparameters on HPC Systems, Oral presentation
- **IEEE/ACM Supercomputing (SC)** in Denver, CO 2023
- **IEEE International Conference on e-Science** in Limassol, Cyprus 2023
 - Towards Lightweight Data Integration using Multi-workflow Provenance and Data Observability, Oral presentation
- **Brazilian Symposium on Databases (SBBD)** in Rio de Janeiro, RJ (virtual) 2021
 - User Steering Support in Large-Scale Workflows, Oral presentation , link
- **Federal Fluminense University (UFF) Computer Science Seminars** in Rio de Janeiro, RJ (virtual) 2021
 - A Knowledge-centric Approach to Support Large-scale AI Systems, Invited talk (Portuguese) , link
- **SIAM Conference on Computational Science and Engineering** in Forth Worth, TX (virtual) 2021
 - AI4Seismic: An AI-Driven Platform to Accelerate Geological Discoveries, Invited talk, Highlighted by the SIAM press , link
 - Workflow Provenance in the Lifecycle of Scientific Machine Learning, Oral presentation , link
- **ACM International Conference on Management of Data (SIGMOD)** in Portland, OR (virtual) 2020
- **Brazilian Symposium on Databases (SBBD)** in Rio (virtual) 2020
- **High-Performance Data Science workshop** in Rio (virtual) 2020
- **Computational Science and Engineering Seminar at COPPE/UFRJ** in Rio (virtual) 2020
 - Workflow Provenance in the Lifecycle of Scientific Machine Learning, Invited talk
- **Open Subsurface Data Universe Development Workshop** in Houston, TX 2020
- **Open Subsurface Data Universe Development Workshop** in Houston, TX 2019
- **IEEE/ACM Supercomputing (SC)** in Denver, CO 2019
 - Workflows in Support of Large-scale Science (WORKS)*
 - Provenance Data in the Machine Learning Lifecycle in Computational Science and Engineering, Oral presentation
- **Scientific Data Analysis using Data-intensive Scalable Computing Workshop** in Rio de Janeiro, Brazil 2019
 - Provenance Data in the Machine Learning Lifecycle in Computational Science and Engineering, Invited talk
- **Open Subsurface Data Universe F2F Meeting** in Houston, TX 2019
- **IEEE International Conference on e-Science** in San Diego, CA 2019
 - Efficient Runtime Capture of Multiworkflow Data using Provenance, Oral presentation
- **Inria Talks** in Montpellier, France 2019
 - Providing Online Data Analytical Support for Humans in the Loop of Computational Science and Engineering Applications, Invited talk

- **IBM Regional Technical Exchange** in Rio de Janeiro, Brazil 2019
- **Provenance Week** in London, UK 2018
 - International Provenance and Annotation Workshop (IPAW)*
 - Provenance of Dynamic Adaptations in User-steered Dataflows, Oral presentation
 - Capturing Provenance for Runtime Data Analysis in Computational Science and Engineering Applications, Poster presentation
 - Computational Reproducibility Workshop*
 - Provenance of Dynamic Adaptations in User-steered Dataflows, Oral presentation
- **International Conference on Very Large Databases (VLDB)** in Rio de Janeiro, Brazil 2018
 - Latin American Data Science Workshop*
 - Tracking Hyperparameter Tuning in Deep Learning Training, Oral presentation
- **Brazilian Symposium on Databases (SBBD)** in Rio de Janeiro, Brazil 2018
- **Brazilian Symposium on Databases (SBBD)** in Uberlandia, Brazil 2017
 - Spark Scalability Analysis in a Scientific Workflow, Oral presentation
 - Controlling the Parallel Execution of Workflows Relying on a Distributed Database, Oral presentation
- **Federal University of Uberlandia, Brazil** in Uberlandia, Brazil 2017
 - Kubernetes, Invited talk
- **Smart City Cloud Hackathon OpenStack Rio** in Rio de Janeiro, Brazil 2017
- **Computer Science Week at UFRJ** in Rio de Janeiro, Brazil 2017
 - Kubernetes, Oral presentation
- **Brazilian Conference on Artificial Intelligence (BRACIS)** in Recife, Brazil 2017
 - Graph Analytics with Spark, Tutorial
- **IEEE/ACM Supercomputing (SC)** in Salt Lake City, UT 2016
 - Workflows in Support of Large-scale Science (WORKS)*
 - Online Input Data Reduction in Scientific Workflows, Oral presentation
- **ASE BigData/SocialCom/CyberSecurity** in Stanford University, Menlo Park, CA 2014
 - Linked open data publication strategies: Application in networking performance measurement data, poster presentation

Badges and Certifications

- **Machine Learning Specialist Professional** Course duration: 73h — 2022
Exploratory Data Analysis, Regression, Classification, Deep Learning, Reinforcement Learning, Unsupervised Learning, Time Series and Survival Analysis, AI Ethics and Explainability
- **Trustworthy AI and AI Ethics** Course duration: 3.5h — 2022
- **Enterprise Design Thinking Practitioner** 2022
- **LinkedIn Skill Assessment: Python, MySQL, Linux, T-SQL, NoSQL**

Scientific Community Service

- IEEE Transactions on Parallel and Distributed Systems - Reviewer
- Future Generation Computer Systems - Reviewer
- Concurrency Computation Practice and Experience - Reviewer
- Journal of Parallel and Distributed Computing - Reviewer
- The Very Large Databases (VLDB) Journal - Reviewer
- IEEE Transactions on Big Data - Reviewer
- Journal of Cloud Computing - Reviewer
- Computer Physics Communications - Reviewer
- Discover Data - Reviewer
- Frontiers in High Performance Computing - Reviewer and Editorial Board

- International Workshop on AI Principles in Science Communication (AISC) - PC
- International Symposium on Computer Architecture and High Performance Computing (SBACPAD'25) - PC
- Workflows in Distributed Environments (WiDE'24) - PC
- IEEE/ACM Supercomputing (SC'24) - PC
- IEEE International Conference on e-Science (eScience'23) - Session Chair, PC
- Workflows in Support of Large-Scale Science (WORKS'20, 21, 23, 24, 25) - PC
- Brazilian Workshop on Database and Artificial Intelligence Integration - PC
- Brazilian Symposium on Databases (SBB'D'20, 23, 24, 25) - Session Chair, PC
- Innovation Summit on Information Systems (at SBSI'19,20) - PC

Languages

- **English** - Full proficiency
 - Missouri State University, U.S. Duration: 150h — Jun 2012 – Aug 2012
Scientific English for Graduate Students
 - Cultura Inglesa (English Culture), Rio de Janeiro, Brazil 2001 – 2009
- **Portuguese** - Native
- **Spanish** - Fluent reading, intermediate speaking and understanding, limited writing

All Publications and Patents

Journal Articles

- [J1] M. Dorier, A. Gueroudji, V. Hayot-Sasson, H. Nguyen, S. Ockerman, **R. Souza**, T. Bicer, H. Pan, P. Carns, K. Chard, "Toward a persistent event-streaming system for high-performance computing applications," *Frontiers in High Performance Computing*, vol. 3, 2025. DOI: 10.3389/fhpcp.2025.1638203. [Online]. Available: <https://www.frontiersin.org/journals/high-performance-computing/articles/10.3389/fhpcp.2025.1638203/abstract>.
- [J2] D. Bard, K. Chard, S. Witt, I. T. Foster, C. Goble, W. Godoy, J. Gustafsson, U.-U. Haus, S. Hudson, L. Los, **R. Souza**, "Workflows community summit 2024: Future trends and challenges in scientific workflows," *Distributed, Parallel, and Cluster Computing (cs.DC)*, 2024. DOI: <https://doi.org/10.48550/arXiv.2410.14943>. [Online]. Available: <https://arxiv.org/abs/2410.14943>.
- [J3] L. G. Azevedo, **R. Souza**, E. F. d. S. Soares, R. M. Thiago, J. C. C. Tesolin, A. C. Oliveira, M. F. Moreno, "A polystore architecture using knowledge graphs to support queries on heterogeneous data stores," *arXiv preprint Databases (cs.DB)*, 2023. DOI: 10.48550/arXiv.2308.03584. [Online]. Available: <https://arxiv.org/abs/2308.03584>.
- [J4] **R. Souza**, L. G. Azevedo, V. Lourenço, E. Soares, R. Thiago, R. Brandão, D. Civitarese, E. Vital Brazil, M. Moreno, P. Valduriez, M. Mattoso, R. Cerqueira, M. A. S. Netto, "Workflow provenance in the lifecycle of scientific machine learning," *Concurrency and Computation: Practice and Experience*, vol. e6544, pp. 1–21, 2021. [Online]. Available: <https://doi.org/10.1002/cpe.6544>.
- [J5] R. F. Silva, R. M. Badia, V. Bala, D. Bard, P.-T. Bremer, I. Buckley, S. Caino-Lores, K. Chard, C. Goble, S. Jha, ... **R. Souza**, et al. "Workflows community summit 2022: A roadmap revolution," *arXiv preprint Distributed, Parallel, and Cluster Computing (cs.DC)*, 2023. DOI: 10.48550/arXiv.2304.00019. [Online]. Available: <https://arxiv.org/abs/2304.00019>.
- [J6] **R. Souza**, V. Silva, A. A. B. Lima, D. Oliveira, P. Valduriez, M. Mattoso, "Distributed in-memory data management for workflow executions," *PeerJ Computer Science*, vol. 7, pp. 1–30, 2021. DOI: 10.7717/peerj-cs.527. [Online]. Available: <https://peerj.com/articles/cs-527/>.
- [J7] R. F. Silva, H. Casanova, K. Chard, ... **R. Souza**, et al. "Workflows community summit: Advancing the state-of-the-art of scientific workflows management systems research and development," 2021, pp. 1–24. [Online]. Available: <https://arxiv.org/abs/2106.05177>.

- [J8] L. G. Azevedo, **R. Souza**, R. Brandão, V. N. Lourenço, M. Costalonga, M. Machado, M. Moreno, R. Cerqueira, "Adding hyperknowledge-enabled data lineage to a machine learning workflow management system for oil and gas," *First Break*, vol. 38, no. 7, pp. 89–93, 2020. DOI: 10.3997/1365-2397.fb2020055.
- [J9] **R. Souza**, V. Silva, A. L. G. A. Coutinho, P. Valduriez, M. Mattoso, "Data reduction in scientific workflows using provenance monitoring and user steering," *Future Generation Computer Systems*, vol. 110, pp. 481–501, 2017, ISSN: 0167-739X. DOI: 10.1016/j.future.2017.11.028.
- [J10] **R. Souza**, V. Silva, J. J. Camata, A. L. G. A. Coutinho, P. Valduriez, M. Mattoso, "Keeping track of user steering actions in dynamic workflows," *Future Generation Computer Systems*, vol. 99, pp. 624–643, 2019, ISSN: 0167-739X. DOI: 10.1016/j.future.2019.05.011. [Online]. Available: <https://doi.org/10.1016/j.future.2019.05.011>.
- [J11] V. Silva, L. Neves, **R. Souza**, A. L. G. A. Coutinho, D. Oliveira, M. Mattoso, "Adding domain data to code profiling tools to debug workflow parallel execution," *Future Generation Computer Systems*, pp. 624–643, 2018, ISSN: 0167-739X. DOI: 10.1016/j.future.2018.05.078.
- [J12] M. G. Bayser, P. Cavalin, **R. Souza**, A. Braz, H. Candello, C. Pinhanez, J.-P. Briot, "A hybrid architecture for multi-party conversational systems," *arXiv preprint Computation and Language (cs.CL)*, pp. 1–40, 2017. [Online]. Available: <https://arxiv.org/abs/1705.01214>.

Conference and Workshop Papers.....

- [C1] D. Rosendo, S. DeWitt, **R. Souza**, P. Austria, T. Ghosal, M. McDonnell, R. Miller, T. J. Skluzacek, J. Haley, B. Turcksin, "Ai agents for enabling autonomous experiments at ornl's hpc and manufacturing user facilities," in *Extreme-Scale Experiment-in-the-Loop Computing (XLOOP) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, St Louis, MO, USA: ACM, 2025.
- [C2] **R. Souza**, T. Poteet, B. Etz, D. Rosendo, A. Gueroudji, "LLM agents for interactive workflow provenance: Reference architecture and evaluation methodology," in *Workflows in Support of Large-Scale Science (WORKS) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, St Louis, MO, USA: ACM, 2025. DOI: 10.1145/3731599.3767582.
- [C3] W. Shin, **R. Souza**, D. Rosendo, F. Suter, F. Wang, P. Balaprakash, R. F. Silva, "The (r) evolution of scientific workflows in the agentic ai era: Towards autonomous science," in *Workflows in Support of Large-Scale Science (WORKS) co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, St Louis, MO, USA: ACM, 2025.
- [C4] **R. Souza**, A. Gueroudji, S. DeWitt, D. Rosendo, T. Ghosal, R. Ross, P. Balaprakash, R. F. Silva, "Prov-agent: Unified provenance for tracking AI agent interactions in agentic workflows," in *IEEE International Conference on e-Science*, Chicago, USA: IEEE, 2025.
- [C5] A. Gueroudji, T. Mallick, **R. Souza**, R. F. Silva, R. Ross, M. Dorier, P. Carns, K. Chard, I. Foster, "Controla: Agentic workflow control mechanisms for reliable science," in *IEEE International Conference on e-Science*, Chicago, USA: IEEE, 2025.
- [C6] **R. Souza**, S. Caino-Lores, M. Coletti, T. J. Skluzacek, A. Costan, F. Suter, M. Mattoso, R. F. Silva, "Workflow provenance in the computing continuum for responsible, trustworthy, and energy-efficient AI," in *IEEE International Conference on e-Science*, Osaka, Japan: IEEE, 2024. DOI: <https://doi.org/10.1109/e-Science62913.2024.10678731>.
- [C7] M. Coletti, **R. Souza**, T. J. Skluzacek, F. Suter, R. F. Silva, "Integrating evolutionary algorithms with distributed deep learning for optimizing hyperparameters on HPC system," in *Workflows in Support of Large-Scale Science (WORKS) workshop co-located with the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, Atlanta, USA: IEEE, 2024.
- [C8] R. F. Silva, W. Shin, F. Suter, A. Gainaru, **R. Souza**, D. Dietz, S. Jha, "Eco-Driven AI-HPC: Optimizing energy efficiency in distributed scientific workflows," in *Energy-Efficient Computing for Science Workshop*, Bethesda, MD, USA, 2024.

- [C9] T. J. Skluzacek, **R. Souza**, M. Coletti, F. Suter, R. F. Silva, "Towards cross-facility workflows orchestration through distributed automation," in *Practice and Experience in Advanced Research Computing (PEARC 24)*, Providence, RI, USA: Association for Computing Machinery, 2024. DOI: 10.1145/3626203.3670606. [Online]. Available: <https://doi.org/10.1145/3626203.3670606>.
- [C10] R. F. Silva, K. Maheshwari, T. Skluzacek, **R. Souza**, S. Wilkinson, "Advancing computational earth sciences: Innovations and challenges in scientific hpc workflows," in *European Geosciences Union (EGU)*, 2024.
- [C11] L. G. Azevedo, **R. Souza**, E. Soares, R. M. Thiago, J. C. C. Tesolin, A. C. C. M. Oliveira, M. F. Moreno, "HKPoly: A polystore architecture to support data linkage and queries on distributed and heterogeneous data," in *Proceedings of the 20th Brazilian Symposium on Information Systems (SBSI)*, ser. SBSI '24, New York, NY, USA: Association for Computing Machinery, 2024, ISBN: 9798400709968. DOI: 10.1145/3658271.3658322. [Online]. Available: <https://doi.org/10.1145/3658271.3658322>.
- [C12] **R. Souza**, T. J. Skluzacek, S. R. Wilkinson, M. Ziatdinov, R. F. Silva, "Towards lightweight data integration using multi-workflow provenance and data observability," in *IEEE International Conference on e-Science*, 2023. DOI: 10.1109/e-Science58273.2023.10254822. [Online]. Available: <https://doi.org/10.1109/e-Science58273.2023.10254822>.
- [C13] D. Rosendo, M. Mattoso, A. Costan, **R. Souza**, D. Pina, P. Valduriez, G. Antoniu, "ProvLight: Efficient workflow provenance capture on the edge-to-cloud continuum," in *IEEE International Conference on Cluster Computing*, 2023. DOI: 10.1109/CLUSTER52292.2023.00026. [Online]. Available: <https://www.computer.org/csdl/proceedings-article/cluster/2023/079200a221/1SfUrCnjgAM>.
- [C14] R. L. Cunha, L. V. Real, **R. Souza**, B. Silva, M. A. Netto, "Context-aware execution migration tool for data science jupyter notebooks on hybrid clouds," in *IEEE International Conference on e-Science*, 2021. DOI: 10.1109/eScience51609.2021.00013.
- [C15] L. Azevedo, **R. Souza**, E. Soares, R. Thiago, A. Oliveira, M. Moreno, "Supporting polystore queries using provenance in a hyperknowledge graph," in *International Semantic Web Conference (ISWC)*, 2021, pp. 1–4.
- [C16] **R. Souza**, "User steering support in large-scale workflows," in *PhD Thesis Contest: Brazilian Symposium on Databases (SBBDD)*, 2021.
- [C17] E. Soares, **R. Souza**, R. Thiago, M. Machado, L. Azevedo, "A recommender for choosing data systems based on application profiling and benchmarking," in *Brazilian Symposium on Databases (SBBDD)*, 2021, pp. 265–270.
- [C18] R. Brandão, V. Lourenço, M. Machado, L. Azevedo, M. Cardoso, **R. Souza**, G. Lima, R. Cerqueira, M. Moreno, "Cycle orchestrator: A knowledge-based approach for structuring cyclic ml pipelines in the o&g industry," in *International Semantic Web Conference (ISWC)*, 2020.
- [C19] R. Brandão, V. Lourenço, M. Machado, L. Azevedo, M. Cardoso, **R. Souza**, G. Lima, R. Cerqueira, M. Moreno, "A knowledge-based approach for structuring cyclic workflows," in *International Semantic Web Conference (ISWC)*, 2020.
- [C20] **R. Souza**, J. Camata, M. Mattoso, A. Coutinho, "Runtime steering of parallel cfd simulations," in *International Conference on Parallel Computational Fluid Dynamics*, 2020.
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